

Requirements

This document lists the dependencies and versions required to compile or run the PowerLLM config-tuning codebase.

Runtime Constraints

Python version: `>=3.11,<3.14`

Required Third-Party Packages

- beautifulsoup4
- cn2an
- docling>=2.0.0
- docling-core
- fastapi
- google-genai
- httpx
- imagehash
- langchain>=1.2.0
- langchain-chroma
- langchain-community>=0.3.1
- langchain-core>=1.2.0
- langchain-google-genai
- langchain-openai
- langchain-text-splitters
- langgraph
- markdown
- openai
- opencv-python-headless
- optuna
- optuna-dashboard
- paddleocr
- pillow
- protobuf>=3.19,<4
- pydantic>=2.0
- python-dotenv
- python-multipart
- rank-bm25
- requests

- sentence-transformers
- sqlite-vss>=0.1.2
- typer
- typing-extensions
- unstructured
- uvicorn

Notes

The package **requests** is included because it is imported directly by the runtime source code.

The module **pathlib** is not listed as an install requirement here because it is part of the Python standard library in Python 3.11 and newer.

This PDF is intended as the deliverable dependency summary and does not modify the project source files.